

Measuring and Managing Risk at Scale: Index Insurance and Aflatoxin Early-Warning

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September 15, 2025

Agenda



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| ① Motivation & Context | ⑤ Prediction Overview |
| ② Questions & Contributions | ⑥ Evaluation Plan & Metrics |
| ③ Methods Overview | ⑦ Results (SZ → MZ) |
| ④ Our Method | ⑧ Implications & Takeaways |

Our method outperforms current methods with **simple** contracts

Main result

Our method captures **52%** of perfect-insurance welfare gains and outperforms current methods. **Joint (multi-zone)** optimization increases this to **59%**.

Why it works. Portfolio-risk pricing allows for more coverage, joint design shifts capital to where marginal utility per unit of portfolio risk is highest.

Three takeaways

- 1 Index insurance **can be effectively optimized** while keeping contracts **simple**.
- 2 **Joint** (multi-zone) optimization delivers **more welfare gains**.
- 3 In noisy, short datasets typical of many developing countries, **simple** piecewise-linear contracts **outperform more complex** ones (Less is more Xie, Ma, and Xin (2024)).

Costly verification + covariate shocks make indemnity hard to scale

- Farmers face a lot of risk, and the lack of risk management tools forces them to use coping strategies that hurt their long term welfare.
- Traditional insurance is prohibitively costly in most developing countries due to lack of data and high verification costs.
- Moral hazard, adverse selection, and the presence of large covariate shocks make the problem of agricultural insurance especially hard.

Index insurance reduces verification & moral hazard — but real-world performance hinges on design & pricing

- **What it is.** Type of insurance where payouts are based on predicted loss (index) rather than manual loss verification.
- **Why it helps.** Automating payouts reduces loss verification costs and prevents moral hazard; contracts are easy to scale.
- **Problems.** Programs operate at scale in India, Mexico, Kenya, Malawi, Tanzania, and other countries, but demand is often low and products usually need subsidizing.
- **Basis risk.** Since it is based on predicted loss, there is the possibility that a farmer suffers loss but doesn't receive a payout.

II is typically subsidized, often bundled with credit, delivered via PPPs

- Index insurance is a named-peril insurance and it is often tied to access to credit.
- Offered at different levels (micro level, meso level, and macro level)
- Often based on public-private partnerships. Research institutions usually design the product, and insurance companies (or governments) price/sell it.



Questions and Contributions

What we ask and what we contribute

Research Questions

- How can we **design simple (piecewise-linear)** index insurance contracts that maximize farmer welfare under **risk-adjusted pricing**?
- What do we gain by **simultaneously designing contracts** across zones when capital is priced on the portfolio?

Contributions

- A **utility-maximizing** program with premiums that incorporate the cost of capital.
- A **multi-zone** program to simultaneously design contracts for multiple zones and a simple proof that it improves over independent zone design.
- An evaluation using a novel dataset of Thai rice farmers.

Index Insurance Literature

- **Impact of Index Insurance:** Overall, there is evidence that index insurance reduces reliance on detrimental risk-coping strategies, increases investment, and leads to riskier, but more profitable production decisions (Jensen and Barrett (2017); Cole et al. (2013); Mobarak and Rosenzweig (2013); Karlan et al. (2014)).
- **Demand for Index Insurance:** Demand for index insurance tends to be low and highly price sensitive (Jensen and Barrett (2017); Cole et al. (2013); Cai, De Janvry, and Sadoulet (2020), Casaburi and Willis (2018)).
- **Design of Index Insurance:** There has been relatively little research studying the design of index insurance. The method developed by Chantarat et al. (2013) is the most commonly used in academic publications (Jensen, Stoeffler, et al. (2019); Flatnes, Carter, and Mercovich (2018)). More recently, Chen et al. (2023) used a NN approach for contract design.



Methods Overview

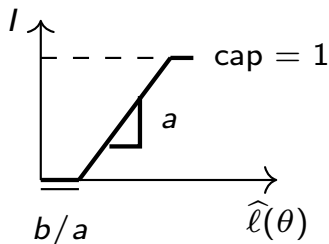
Piecewise-linear contracts: understandable and adjustable

Contract form

$$I(\theta) = \min\{\max\{0, a\widehat{\ell}(\theta) - b\}, 1\}$$

where $\widehat{\ell}(\theta)$ is predicted loss, $a \geq 0$ is the slope, $b \geq 0$ is the strike/intercept, and payouts are capped at 1 (normalized).

- *Deductible-like*: theoretically optimal (Arrow (1974)).
- *Interpretable*: simple contracts are easier to explain to farmers, this is important for take up (Ahmed, McIntosh, and Sarris (2020)).
- *Popular*: this contract structure is the most commonly used by practitioners (Gamage (2021)).



0 below strike; slope a until cap; flat thereafter.

Baseline Method: Chantarat et al. (2013)

Developed by Chantarat et al. (2013), used in Kenya Index Based Livestock Insurance (IBLI) program. Consist of two steps:

- 1 Train model to predict loss
- 2 Choose strike value, b , that maximizes correlation between payouts and losses

$$I(\theta) = \min\{\max\{0, \hat{\ell}(\theta) - b\}, 1\}$$

Pros: simple, interpretable

Cons: can't incorporate constraints, doesn't take into account premium

NN Approach: Chen et al. (2023)

Train neural network to solve the following optimization problem:

$$\begin{aligned} \max \quad & \mathbb{E} [u(w_0 + 1 - \ell + I - \pi)] \\ \text{s.t.} \quad & \pi = \lambda \mathbb{E} [I] \\ & \pi \leq \bar{\pi} \end{aligned}$$

Use CARA utility function:

$$\begin{aligned} u(w) &= \frac{-1}{\alpha} e^{-\alpha w} \\ \alpha &> 0 \end{aligned}$$

Pros: Directly maximizes utility, admits price constraints.

Cons: Not interpretable. hard to adjust, large data requirements, simplistic premium

We predict losses, then optimize simple contracts under risk-adjusted pricing

Our approach is most similar to the baseline method. We predict-then-design. The steps are:

- 1 Train a traditional ML model to predict loss
- 2 Use convex optimization model to design simple, utility-maximizing contracts

This yields simple and interpretable contracts.

Method comparison: objective, pricing, contract class, interpretability

	Chantararat (Baseline)	Chen (NN)	Ours (VMX)
Objective	Maximize correlation of payouts with losses	Maximize expected utility	Maximize expected utility
Pricing	Not incorporated	$\pi = \lambda \mathbb{E}[I]$	$\pi = \mathbb{E}[I] + c_{\kappa}(\text{CVaR}_{1-\varepsilon}(I) - \mathbb{E}[I])$
Contract class	Piecewise-linear	Non-parametric	Piecewise-linear
Interpretability	High	Low	High



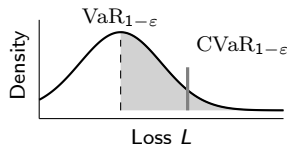
Our Method

CVaR: definition, coherence, and how it enters pricing

Definitions (for loss L , CDF F_L , confidence $\epsilon \in (0, 1)$)

$$\text{VaR}_{1-\epsilon}(L) := \inf\{\ell \in \mathbb{R} : F_L(\ell) \geq 1 - \epsilon\},$$

$$\text{CVaR}_{1-\epsilon}(L) := \mathbb{E}[L \mid L \geq \text{VaR}_{1-\epsilon}(L)]$$



Risk-adjusted premium (normalized payouts I)

$$\pi_z = \mathbb{E}[I_z] + c_\kappa \frac{\beta_z}{s_z} \underbrace{\left(\text{CVaR}_{1-\epsilon}\left(\sum_z I_z\right) - \mathbb{E}\left[\sum_z I_z\right] \right)}_{\text{Portfolio required capital}}$$

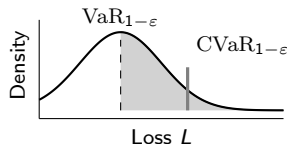
Premium definition from *Risk Modeling for Appraising Named Peril Index Insurance Products: A Guide for Practitioners* Mapfumo, Groenendaal, and Dugger (2017).

CVaR: definition, coherence, and how it enters pricing

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Why CVaR?

- Coherent risk measure: monotonicity, translation invariance, positive homogeneity, **subadditivity**; also convex.
- Widely used in practice and regulation (e.g., solvency regimes in Switzerland, Canada, USA).

Premium definition from *Risk Modeling for Appraising Named Peril Index Insurance Products: A Guide for Practitioners* Mapfumo, Groenendaal, and Dugger (2017).

Premium Principles

A premium principle is a functional that maps an insurance contract to a price (premium). There are several premium principles that are common in the actuarial literature, and our method is compatible with most of them:

Description	Definition
Expected value	$(1 + \alpha)\mathbb{E}[I]$
Standard deviation	$\mathbb{E}[I] + \alpha\sigma(I)$
Variance	$\mathbb{E}[I] + \alpha\sigma^2(I)$
Exponential	$\frac{1}{\alpha} \log \mathbb{E}[e^{\alpha I}]$
Dutch	$\mathbb{E}[I] + \beta\mathbb{E}[(I - \alpha\mathbb{E}[I])_+]$

Pricing and Design

- The contracts are **priced using portfolio risk**¹, but the contract for each zone is **designed independently**, as if other zones didn't exist.
- The final premiums paid by farmers are still based on **portfolio capital**:

$$\pi_z = \mathbb{E}[I_z] + c_k \frac{\beta_z}{s_z} \left(\text{CVaR}_{1-\epsilon} \left(\sum_z s_z I_z \right) - \mathbb{E} \left[\sum_z s_z I_z \right] \right)$$

- However, since contracts are designed on zone at a time, portfolio risk isn't observed and a single-zone variant of the premium is used:

$$\pi_z = \mathbb{E}[I_z] + c_k (\text{CVaR}_{1-\epsilon}(I_z) - \mathbb{E}[I_z])$$

¹Mapfumo, Groenendaal, and Dugger (2017)

Why index insurance uses single-zone design while actuarial pricing is portfolio-based

The apparent disconnect. Portfolio-based risk pricing is standard in actuarial practice, yet the index insurance literature often optimizes *one zone at a time*.

Why this happens.

- 1 **Separate literatures.** Index insurance is newer and separate from actuarial literature. It's mostly focused on feasibility and demand. Premium principles are often simplified ($\pi = \lambda \mathbb{E}[I]$), with limited attention to capital/pricing.
- 2 **Technical barriers.** Optimization approach had been previously attempted but abandoned because the resulting problem was non-convex, and practitioners usually didn't have an engineering background.
- 3 **Institutional split.** Public-private model: researchers/NGOs *design* products; insurers/reinsurers later *price* them on the portfolio. The separation reinforced SZ-focused research.

Idealized Model

$$\max_{a,b,\pi} \mathbb{E}[U(w_0 + 1 - \ell - \pi + I(\theta))] \quad (1)$$

$$\text{s.t. } I(\theta) = \min \left\{ \max \left\{ 0, a\hat{\ell}(\theta) - b \right\}, 1 \right\} \quad (2)$$

$$\pi = \mathbb{E}[I(\theta)] + c_{\kappa} [\text{CVaR}_{1-\epsilon_K}(I(\theta)) - \mathbb{E}[I(\theta)]] \quad (3)$$

$$\pi \leq \bar{\pi}. \quad (4)$$

Idealized Model

$$\max_{a,b,\pi} \mathbb{E}[U(w_0 + 1 - \ell - \pi + I(\theta))] \quad (5)$$

$$\text{s.t. } I(\theta) = \min \left\{ \max\{0, a\hat{\ell}(\theta) - b\}, 1 \right\} \quad (6)$$

$$\pi = \mathbb{E}[I(\theta)] + c_\kappa [\text{CVaR}_{1-\epsilon_K}(I(\theta)) - \mathbb{E}[I(\theta)]] \quad (7)$$

$$\pi \leq \bar{\pi}. \quad (8)$$

Notation

w_0 : initial wealth

ℓ : realized loss

$\hat{\ell}$: predicted loss

$I(\theta)$: insurance payout

π : premium paid by the farmer.

Convex Relaxation

$$\begin{aligned} \max_{a,b,\pi} \quad & \mathbb{E} \left[U \left(w_0 + 1 - \ell - \pi + \underline{I(\theta)} \right) \right] \\ \text{s.t.} \quad & \overline{I(\theta)} = \max \left\{ 0, a\hat{\ell}(\theta) - b \right\} \\ & \underline{I(\theta)} = \min \left\{ a\hat{\ell}(\theta) - b, 1 \right\} \\ & \pi = (1 - c_\kappa) \mathbb{E} \left[\overline{I(\theta)} \right] + c_\kappa \text{CVaR}_{1-\epsilon_K} \left(\overline{I(\theta)} \right) \\ & \pi \leq \overline{\pi}. \end{aligned}$$

CP Formulation: Rockafellar and Uryasev (2002)

$$\begin{aligned} \max_{a,b,\{\gamma_K^j\},\{\alpha^j\},\{\omega^j\},t_K,K} \quad & \frac{1}{N} \sum_j U(w_0 + 1 - \ell^j - \pi + \omega^j) \\ \text{s.t.} \quad & \pi = \frac{(1 - c_K)}{N} \sum_j \alpha^j + c_K K \\ & t_K + \frac{1}{\epsilon_K} \sum_j p^j \gamma_K^j \leq K \\ & \gamma_K^j \geq \alpha^j - t_K, \forall j \\ & \gamma_K^j \geq 0, \forall j \\ & \alpha^j \geq a \hat{\ell}(\theta^j) + b, \forall j \\ & \alpha^j \geq 0, \forall j \\ & \omega^j \leq a \hat{\ell}(\theta^j) + b, \forall j \\ & \omega^j \leq 1, \forall j \\ & \pi \leq \bar{\pi}. \end{aligned}$$



Prediction Overview

Prediction: Problem Overview

- For each insured area we have several time-series of different weather variables, sampled at potentially different frequencies:

$$\begin{bmatrix} x_{i,1} & \dots & x_{i,16} & \dots & x_{i,176} \\ y_{i,1} & y_{i,2} & \dots & y_{i,16} & \dots & y_{i,176} & \dots & y_{i,180} \\ z_{i,1} & z_{i,2} & \dots & z_{i,16} & \dots & z_{i,176} & \dots & z_{i,180} \end{bmatrix}$$

- These weather variables are rainfall, temperature, and evapotranspiration.

Feature Extraction

- Time-series classification (TSC) is different from standard problems because the data points are ordered, and the order matters. The space of potential features is also large, e.g. we can take 3-day dry spells, 5-day dry spells, number of high temperature days, etc.
- Ruiz et al. (2021) compared many different specialized TSC algorithms and they found that the RoCKET algorithm (Dempster, Petitjean, and Webb (2020)) generally performed best.
- We use RoCKET, Catch22 (Lubba et al. (2019)), and raw time-series for feature extraction.

Prediction overview — features, models, validation

- **Preprocessing:** uniform scaling, quantile transformer
- **Feature extraction:** Catch22, RoCKET, raw time-series
- **Prediction algorithm:** Random Forests (RF), Gradient Boosting Regressor (GBR), Elastic Net, LASSO



Evaluation Data, Plan, and Metrics

Data in Index Insurance Research

- One of the biggest barriers in AI research is the lack of granular yield data for LMIC's.
- Most previous studies either use a small dataset from a pilot index insurance program in Kenya (Chantararat et al. (2013); Flatnes, Carter, and Mercovich (2018); Estefania-Salazar et al. (2024)) or use data from a rich country (Chen et al. (2023)).
- I collaborated with the Bank of Thailand and obtained access to a novel dataset of losses from natural disasters in Thailand.

Thai agriculture is rain-fed and smallholder-dominated → high weather exposed

- Characterized by rural, smallholder farmers, with an average farm size of 5.7 acres. Average in sub-Saharan Africa, South Asia, and East Asia is ~ 4 acres.
- 76% of agriculture in Thailand is rain-fed. Globally, this ranges from 60% in South Asia to over 90% in sub-Saharan Africa.
- Consistent exposure to floods and droughts, especially in northeast region. LMIC's generally face more flood and drought risk than richer countries (Winsemius et al. (2018)).
- $\sim 80\%$ of poor households are employed in agriculture, and $\sim 30\%$ of overall Thai labor force is employed in agriculture

Data overview

- **Observations:** 26,938 Tambon–year records (average losses from natural disasters).
- **Geographic unit:** Tambon (sub-district) level across Thailand.
- **Time span:** 2015–2022 (8 years).
- **Target:** Average crop losses from natural disasters.
- **Predictors:** Daily climate data aggregated to the growing season:
 - ▶ Rainfall
 - ▶ Temperature
 - ▶ Evapotranspiration

Source: disaster loss reports (Thailand) combined with gridded climate datasets.

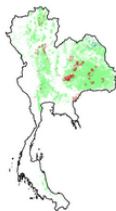
Thai Data: Administrative Data Sources

- **Thailand Department of Agricultural Extension (DOAE):** This dataset contains detailed information on the planting activities of over 3.1 million Thai farmers. Information includes plant date, plot size, rice variety, etc
- **Thai Government Disaster Relief Data:** This data is from the current national disaster relief program. It contains information about the disaster type, losses, and disaster date.

Remote Sensing Data Sources

- **Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS):** daily rainfall data 1981-present
- **MOD21A1 Land Surface Temperature:** daily temperature data, available 2000-present
- **FLDAS Famine Early Warning Systems Network:** monthly evapotranspiration data, available 1992-present

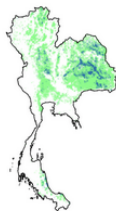
Disaster Activity Map



2015



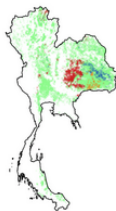
2016



2017



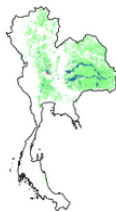
2018



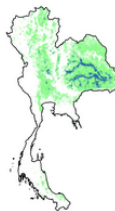
2019



2020



2021



2022

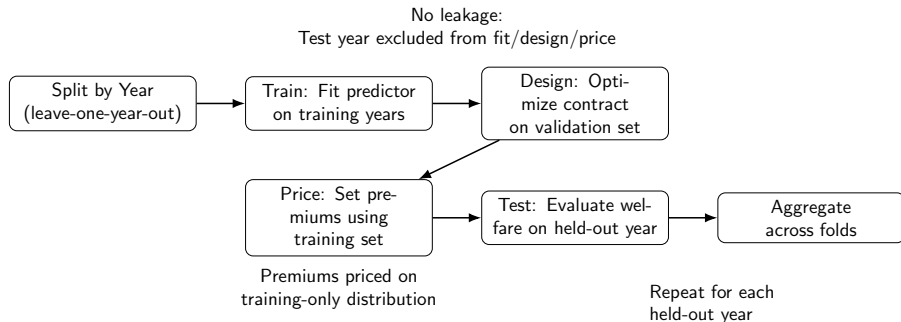
Evaluation plan: leave-one-year-out (train on $y' \neq y$, test on y)

- We use leave one year out cross validation.
- In each CV fold, data from year y is used for testing. Data from $y' \neq y$ is used for insurance design (prediction model training and contract design).
- This gives us out-of-sample predictions for every observation in our data. We then compute utility-based performance metrics across all out-of-sample predictions.
- Out-of-sample predictions used for contract design and evaluation, pricing held constant across methods.

Evaluation Details

- We use the training data and the same method to compute the premium for all methods.
- We modify Chen's method to use the same premium definition as ours.
- We use simulated data to estimate the CVaR when calculating the premium.
- We use a CRRA utility function with $\alpha = 1.5$ (Flatnes, Carter, and Mercovich (2018); Estefania-Salazar et al. (2024); Jensen, Stoeffler, et al. (2019))

Evaluation pipeline (LOYO)



Relative Insurance Benefit (RIB)

- We focus on the Relative Insurance Benefit (RIB) (Kenduiywo et al. (2021); Estefania-Salazar et al. (2024)) as our main metric.
- The RIB of an insurance J is the ratio of the welfare gains of J and the welfare gains of perfect insurance:

$$\text{RIB} = \frac{\text{Welfare gains from insurance } J}{\text{Welfare gains from perfect insurance}}$$

- Interpretation: if RIB of an insurance J is 50%, it yields half as many welfare gains as perfect insurance. We can think of it as meaning that it is 50% as good as perfect insurance.

Relative Insurance Benefit (RIB)

Let w_i^J be the wealth of farmer i under insurance contract J . We denote perfect insurance with PI and no insurance with NI

$$\begin{aligned}CE(J) &: u^{-1}(\mathbb{E}[u(w^J)]) \\ \Delta CE(J) &: = CE(J) - CE(NI) \\ RIB(J) &: \frac{\Delta CE(J)}{\Delta CE(PI)} \\ \text{Utility} &: u(w) = \frac{w^{1-\alpha}}{1-\alpha}\end{aligned}$$



Results (SZ $\rightarrow$ MZ)

SZ: VMX achieves RIB 0.524 vs -0.163 (Chantararat) and -1.909 (NN)

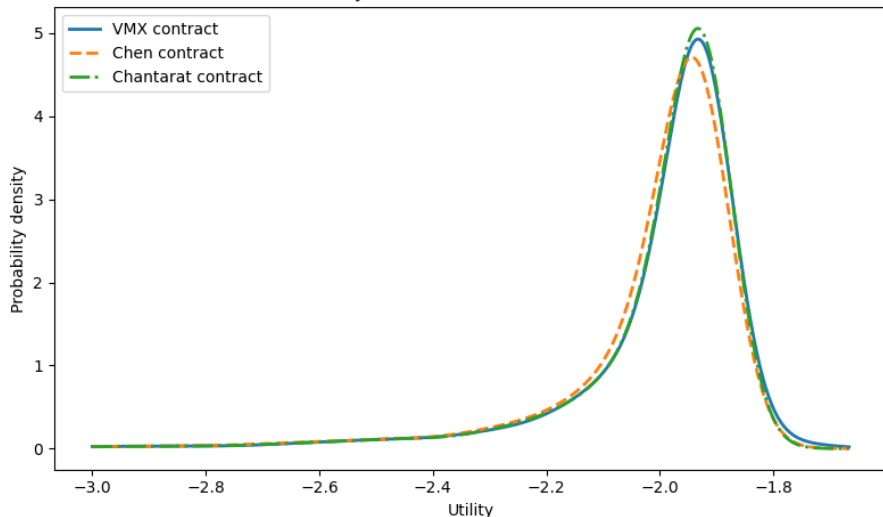
Method	RIB	Premium	Cost ¹	Cost PI ¹
VMX	0.524	0.014	0.017	0.027
Chantararat	-0.163	0.004	0.002	0.027
Chen	-1.909	0.074	0.062	0.027

¹**Cost** is the average payout of the insurance; **Cost PI** is the average payout of perfect insurance.

¹We obtain similar results when we don't use portfolio pricing

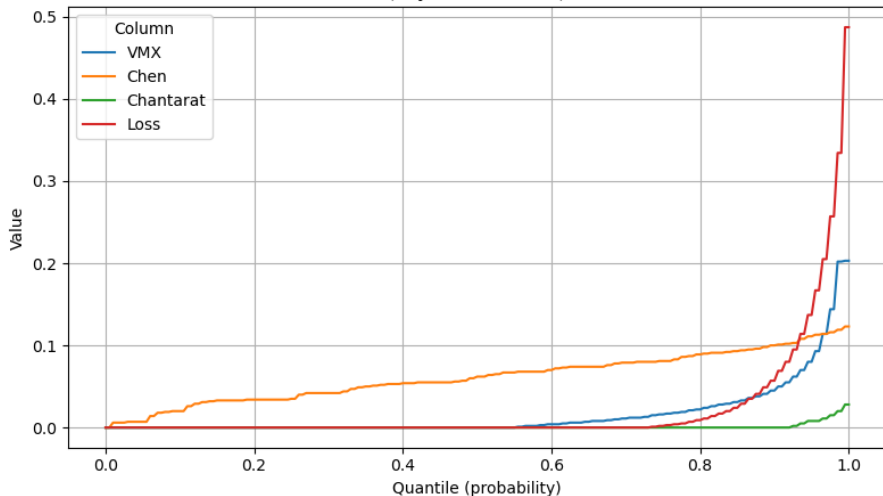
Outperforms in the entire distribution

Utility distributions: VMX vs. Chen



Payout and Loss Quantile Functions

Empirical Quantile Function
(Payouts and Loss)



Why multi-zone?

What we just saw (SZ).

- Our single-zone (SZ) design already improves welfare compared to existing methods.
- Better behavior: more payouts at the tails, less unnecessary payouts.

Why SZ is suboptimal.

- **Diversification.** Just as in finance, the risk of a diversified portfolio is lower than the sum of the risks of each asset. Designing contracts zone by zone ignores this benefit, and overestimates the costs of capital.
- **Misallocation risk.** In portfolio theory, the optimal mix depends on correlations. SZ ignores cross-zone dependence and effectively assumes perfect correlation.

Next step (MZ). Design contracts for all zones **jointly**, so capital is priced on the entire portfolio. This way, we fully use diversification and allocate coverage where it does the most good.

Multiple Zone Case

In the multi-zone case, the problems are coupled through the required capital

$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z s_z I_z(\hat{\ell}_z) \right) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right]$$

The first term will be affected by the correlation of losses across zones, so the optimal contracts will vary depending on this correlation. There are also additional decision variables k_z .

SZ vs MZ — what changes in the optimization problem?

$$\begin{aligned} \max_{a,b,K,\pi} \quad & \mathbb{E} \left[\sum_z s_z U(w_{0,z} + 1 - \ell_z - \pi_z + \underline{l}_z(\theta_z)) \right] \\ \text{s.t.} \quad & \pi_z = \mathbb{E} \left[\overline{l_z(\theta_z)} \right] + \frac{c_\kappa}{s_z} k_z \end{aligned} \quad (9)$$

$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z s_z \overline{l_z(\theta_z)} \right) - \mathbb{E} \left[\sum_z s_z \underline{l_z(\theta_z)} \right] \quad (10)$$

$$K = \sum_z k_z \quad (11)$$

$$\pi_z \leq \overline{\pi_z} \quad (12)$$

$$\overline{l_z(\theta_z)} = \max \left\{ 0, a_z \hat{\ell}_z(\theta_z) - b_z \right\}$$

$$\underline{l_z(\theta_z)} = \min \left\{ a_z \hat{\ell}_z(\theta_z) - b_z, 1 \right\}$$

SZ vs MZ — what changes in the optimization problem?

$$\max_{\{a_z, b_z\}_z, \{\pi_z\}_z, \{k_z\}_z, K} \mathbb{E} \left[\sum_z s_z U(w_{0,z} + 1 - \ell_z - \pi_z + \underline{l}_z(\theta_z)) \right] \quad (13)$$

$$\text{s.t.} \quad \pi_z = \mathbb{E} \left[\overline{l_z(\theta_z)} \right] + \frac{c_\kappa}{s_z} k_z \quad (\text{pricing with capital allocation})$$

$$K = \underbrace{\text{CVaR}_{1-\epsilon} \left(\sum_z s_z \overline{l_z(\theta_z)} \right) - \mathbb{E} \left[\sum_z s_z \underline{l_z(\theta_z)} \right]}_{\text{portfolio capital requirement}}$$

(portfolio risk couples zones)

$$K = \sum_z k_z \quad (\text{shared capital budget})$$

$$\pi_z \leq \overline{\pi_z} \quad (14)$$

$$\overline{l_z(\theta_z)} = \max \left\{ 0, a_z \hat{\ell}_z(\theta_z) - b_z \right\}, \quad (15)$$

$$\underline{l_z(\theta_z)} = \min \left\{ a_z \hat{\ell}_z(\theta_z) - b_z, 1 \right\}. \quad (16)$$

Why $MZ \geq SZ$ sketch

Intuition. Single-zone (SZ) design prices capital *per zone*, ignoring diversification. With a coherent risk measure (CVaR), capital is *subadditive*, so SZ *overestimates* required capital $\Rightarrow$ it solves a *more constrained* problem than MZ.

- 1 **Solve SZ.** Obtain SZ-optimal contracts $\{a_z^{SZ}, b_z^{SZ}\}$ with payouts I_z^{SZ} and premiums π_z^{SZ}
- 2 **Augment to an MZ-feasible solution.** Keep the same contracts (so payouts are unchanged) and compute portfolio capital

$$K^{MZ} = \text{CVaR}_{1-\epsilon}\left(\sum_z s_z I_z^{SZ}\right) - \mathbb{E}\left[\sum_z s_z I_z^{SZ}\right] \leq K^{SZ} \quad (\text{subadditivity}).$$

Pick capital shares $\{k_z^{MZ}\}$ so price caps remain satisfied. $\pi_z^{MZ} \leq \pi_z^{SZ}$ due to lower K .

- 3 **Objective improves.** Payouts are identical and premiums are (weakly) lower, so welfare (expected utility) is weakly higher under this MZ-feasible solution.

Conclusion.

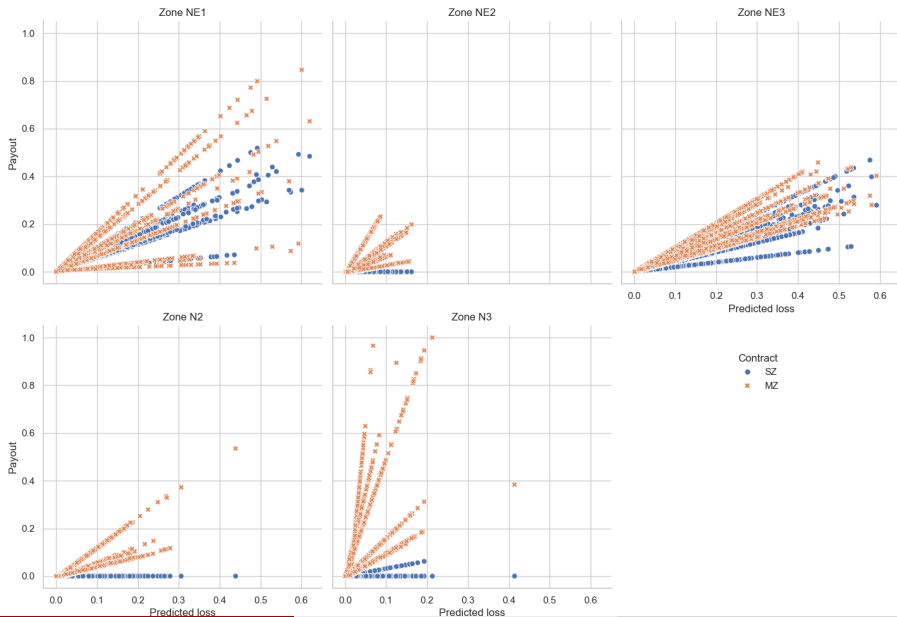
$$P_2^* \geq \sum_z P_1(z)^*$$

(equality only under comonotonic losses).

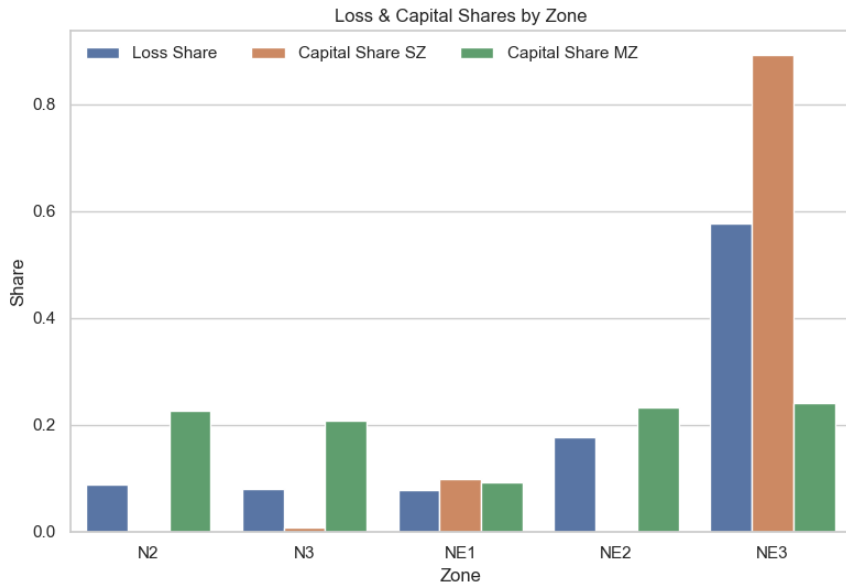
MZ: joint design raises RIB 0.524 $\rightarrow$ 0.592 at realistic premiums

Method	RIB	Premium	Cost	Cost PI
VMX-M	0.592	0.035	0.036	0.027
VMX	0.524	0.014	0.017	0.027
Chantararat	-0.163	0.004	0.002	0.027
Chen	-1.909	0.074	0.062	0.027

MZ contracts tend to be steeper



Capital costs are reallocated to benefit high-loss zones



Utility gains/losses from insurance (relative to no insurance)

$$\Delta U = U(w_0 + 1 - \ell - \pi + I) - U(w_0 + 1 - \ell)$$

	Payout ($I > 0$)	No payout ($I = 0$)
Loss occurs	Gain — Good gain : loss is covered by payout; typically $\Delta U \geq 0$.	Loss — Bad loss : loss with no payout and premium; $\Delta U < 0$.
No loss	Gain : payout despite no loss; $\Delta U > 0$.	Loss : premium cost when no loss; $\Delta U < 0$.

Terminology: We call the first type of utility *gain* a **good gain** (loss with payout). We call the second type of utility *loss* a **bad loss** (loss without payout, plus premium).

Good gains and bad losses: welfare-focused view

Let $\Delta u_i \equiv u(w_i^{\text{ins}}) - u(w_i^0)$.

Metrics we report

$$\text{Good Gain} = \frac{\sum_{i:L_i>0} w_i (\Delta u_i)_+}{\sum_{i:\Delta u_i>0} w_i (\Delta u_i)_+}$$

share of utility gains that occur when there is a loss

$$\text{Bad Loss} = \frac{\sum_{i:\Delta u_i<0 \wedge L_i>0} w_i (\Delta u_i)_+}{\sum_{i:\Delta u_i<0} w_i (\Delta u_i)_+}$$

share of utility losses that occur when there is a loss

$$(x)_+ = \max\{x, 0\}$$

Interpretation

- **Higher** Good Gain $\Rightarrow$ insurance is behaving as intended.
- **Lower** Bad Loss $\Rightarrow$ less basis risk.

Insurance behavior: payout precision and loss recall

Definitions (normalized per plot i with weights w_i)

Let I_i be the (normalized) payout and L_i the realized loss.

$$\textbf{Payout Precision } P = \frac{\sum_i w_i \min\{I_i, L_i\}}{\sum_i w_i I_i} \quad (\text{share of payouts that were valid})$$

$$\textbf{Loss Recall } R = \frac{\sum_i w_i \min\{I_i, L_i\}}{\sum_i w_i L_i} \quad (\text{share of loss dollars that were covered})$$

- P penalizes over-payment; R penalizes under-payment when $L_i > 0$.
- Multi-zone (portfolio) design boosts R by using diversification and capital allocation to make larger payouts.

Insurance behavior results (VMX-M vs VMX and baselines)

Method	RIB	Payout Precision	Loss Recall	Good Gain	Bad Loss
VMX-M	0.597	0.184	0.211	0.390	0.334
VMX	0.524	0.175	0.120	0.335	0.372
Chantararat	-0.222	0.328	0.010	0.616	0.514
Chen	-1.924	0.173	0.298	0.327	0.440
Δ VMX-M – VMX	+0.075	+0.009	+0.091	+0.055	−0.038

- **Multi-zone (VMX-M)** raises welfare (+0.075 RIB) by *covering more true losses* (Recall +0.091) and *reducing under-protected loss states* (Bad Loss ↓ 0.038).
- Chantararat's high “precision” reflects very few payouts overall (low recall); VMX-M improves balance.

Summary of results

Main result

Our method captures about **52%** of the welfare gains of perfect insurance and outperforms current methods. Joint (multi-zone) optimization further increases welfare (*from 0.524 to **0.592** in RIB terms*).

Why it works

- **Diversification lowers capital:** portfolio CVaR is *subadditive*, so required capital is lower on the portfolio than the sum of per-zone capitals, allowing for steeper contracts.
- **Capital reallocation:** with a portfolio risk load, capital shifts toward zones with highest utility gain per unit of portfolio risk.

Takeaways and Next steps

Three takeaways

- ① Index insurance **can be effectively optimized** while keeping contracts **simple**.
- ② **Joint** (multi-zone) optimization delivers **more welfare gains**.
- ③ In noisy, short datasets typical of many developing countries, **simple** piecewise-linear contracts **outperform more complex** ones (Less is more Xie, Ma, and Xin (2024)).

Next Steps

- ① Extend MZ dominance proof to general contract class and more general premium principle.
- ② Run simulation study to understand how performance gap is affected by correlation of losses amongst zones.
- ③ Do an analysis of FOC conditions to understand how correlation and prediction quality affect optimal parameter values.

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NN Approach: Chen et al. (2023)

Train neural network to solve the following optimization problem:

$$\begin{aligned} \max \quad & \mathbb{E}[u(w_0 + 1 - \ell + I - \pi)] \\ \text{s.t.} \quad & \pi = \lambda \mathbb{E}[I] \\ & \pi \leq \bar{\pi} \end{aligned}$$

Use CARA utility function:

$$\begin{aligned} u(w) &= \frac{-1}{\alpha} e^{-\alpha w} \\ \alpha &> 0 \end{aligned}$$

NN Approach: Chen et al. (2023)

Train neural network to solve the following optimization problem:

$$\begin{aligned} \max \quad & \mathbb{E}[u(w_0 + 1 - \ell + I - \pi)] \\ \text{s.t.} \quad & \pi = (1 - c_\kappa) \mathbb{E}[\overline{I(\theta)}] + c_\kappa \text{CVaR}_{1-\epsilon_K}(\overline{I(\theta)}) \\ & \pi \leq \bar{\pi} \end{aligned}$$

Use CRRA utility function:

$$\begin{aligned} u(w) &= \frac{w^{1-\alpha}}{1-\alpha} \\ \alpha &= 1.5 \end{aligned}$$

Estimating portfolio CVaR via copula-based scenario generation

What is a copula (one line). $F_{L_1, \dots, L_Z}(\ell_1, \dots, \ell_Z) = C(F_1(\ell_1), \dots, F_Z(\ell_Z))$. A copula C couples *marginals* F_Z into a joint distribution, letting us model dependence (incl. tail dependence) separately from per-zone loss behavior.

Why here (short/noisy data).

- We need reliable $\text{CVaR}_{1-\varepsilon}$ of the *portfolio* $S = \sum_Z s_Z I_Z$ but historical series are short/noisy.
- Copulas preserve cross-zone dependence while augmenting tails; widely used in finance/actuarial (portfolio aggregation, economic capital, reinsurance pricing, Solvency regimes).

How we use it (pipeline).

- 1 Fit per-zone marginals F_Z (empirical + tail fit or parametric) on training years.
- 2 Fit a dependence model C (e.g., t -copula with correlation Σ and df ν ; Gaussian as a baseline).
- 3 Simulate $u^{(m)} \sim C$; set $L_Z^{(m)} = F_Z^{-1}(u_Z^{(m)})$; compute payouts $I_Z^{(m)}$ and

Idealized CVaR Model

- **Objective:** conditional value at risk of the farmers' loss net of insurance.
- **Constraint 1:** piecewise-linear structure of the contract.
- **Constraint 2:** budget constraint.
- **Constraint 3:** definition of required capital.

$$\min_{a,b,\pi,K} \text{CVaR}_{1-\epsilon}(\ell - I(\theta))$$

$$\text{s.t. } I(\theta) = \min\{(a\hat{\ell}(\theta) + b)^+, P\} \quad (17)$$

$$\mathbb{E}[I(\theta)] + c_\kappa K \leq B \quad (18)$$

$$K = (\text{CVaR}_{1-\epsilon}(I(\theta)) - \mathbb{E}[I(\theta)]) \quad (19)$$

The problem is non-convex, so we need convex approximations

We use the following approximations of $I(\theta)$ to make the problem convex:

$$\overline{I(\theta)} \triangleq \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}$$
$$\underline{I(\theta)} \triangleq \min \{ a\hat{\ell}(\theta) + b, K \}$$

- Note that $\overline{I(\theta)} \geq I(\theta)$ and $\overline{I(\theta)}$ is convex. Conversely, $\underline{I(\theta)} \leq I(\theta)$ and $\underline{I(\theta)}$ is concave.
- We replace $I(\theta)$ with either $\overline{I(\theta)}$ or $\underline{I(\theta)}$ where necessary to obtain conservative and convex approximations.
- We also need approximations or proxies for $E[I(\theta)]$ in constraint . We use $\pi_{SQ} = E[I_{SQ}(\theta)]$, where I_{SQ} is the contract designed using the status quo method, as a proxy for $E[I(\theta)]$ in constraint .